

Andi Syaichul Mubaraq

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Summary

A highly motivated Information Systems and Technology undergraduate at Institut Teknologi Bandung with a strong passion for software engineering, cybersecurity, and data management. Experienced in building cross-platform applications, architecting secure zero-knowledge systems, and implementing data processing algorithms. Eager to leverage technical skills in Java, Python, and React Native to deliver impactful technology solutions and gain practical industry experience in an enterprise IT environment through an internship program.

Experience

Himpunan Mahasiswa Informatika (HMIF) ITB - Staff of Household Division Aug 2025 – Mar 2026

- Managed and maintained the HMIF secretariat facilities and organizational properties, ensuring a clean, safe, and conducive environment for members' daily operations and activities.

Education

Institut Teknologi Bandung - B. Eng. in Information Systems and Technology Aug 2023 - Present

Projects

LinePass — Zero-Knowledge CLI Password Manager 2026

Designed a highly secure CLI credential vault utilizing AES-128-GCM encryption, PBKDF2 key derivation, and Shamir's Secret Sharing combined with Naor-Shamir visual cryptography to securely distribute and recover master keys via pixel-manipulated QR codes.

meowchat — End-to-End Encrypted Messaging Web App 2026

Engineered a containerized full-stack messaging application (React, Node.js, SQLite) featuring a zero-knowledge server architecture by implementing End-to-End Encryption (E2EE) via Web Crypto API and building a custom ECDSA-based JWT authentication system entirely from scratch.

Stego Pixel — Video Steganography Application 2026

Built a Python/PyQt5 desktop GUI application leveraging OpenCV for frame-level pixel manipulation and Matplotlib for comparative histogram analysis to securely embed and extract cryptographic messages within video files using the Least Significant Bit (LSB) method.

Searching - Sorting — Algorithm Visualization Mobile Lab 2026

Developed a cross-platform mobile application utilizing React Native, TypeScript, and Firebase to dynamically visualize complex algorithmic computations, featuring real-time UI state management for animation controls and automated deployment via Expo Application Services.

DIA atau Aku — Supervised Machine Learning & Data Preprocessing 2025

Executed comprehensive data preprocessing pipelines (cleaning, dimensionality reduction, feature selection) using Pandas and NumPy, and co-developed Object-Oriented Decision Tree and K-Nearest Neighbors algorithms from scratch to benchmark against Scikit-learn baselines.

Ruang Pandai — Desktop Tutor Management System 2025

Architected a modular desktop management system applying the Entity-Boundary-Controller (MVC) pattern to strictly decouple Java business logic from the JavaFX UI, utilizing Apache Maven for build automation and complex relational database management.

Spakbor Hills — 2D CLI Farming Game 2025

Programmed the core mechanics and state logic of a 2D CLI farming game in Java using advanced OOP Design Patterns (Singleton, Factory, Template Method), collaborating within a 5-person team using Git and Gradle for version control and continuous integration.

Skills

Languages:	Python, Java, JavaScript, TypeScript, C, C++, SQL.
Technologies:	React Native, React.js, Node.js, Express, Firebase, Expo, SQLite, Git, Docker, OpenCV, NumPy, Pandas, Scikit-learn.
Concepts:	Cryptography, End-to-End Encryption, Object-Oriented Programming, MVC Architecture, Data Structures.